Group Recommender Systems

Video link: https://youtu.be/qSmXtZecNN8

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The content

- Motivation
- Group Recommender Systems
- Basic Steps of Group Recommendation Process
- Further topics in Group Recommendation

Motivation (1)

- Mainstream recommender systems focus on single-user recommendation
- Several decisions are made by groups of users:
  - *Family movie choice*: Movies with family
  - *Family travel selection*: Summer vacation destination
  - *Group dining choice*: Dinner with friends
  - *Workout playlist selection*: Song list at fitness center
  - *Release planning decision*: Next software release



The Image is generated with OpenAI DALL·E, 2026.

Motivation (2)

- Groups are categorized based on how the group was formed [Boratto2010].

Formulation	Explanation of Formulation	Example
 **Established Groups**	Users often choose to be part of the group due to their shared and long-term interests.	• A group of researchers with similar academic interests. • A group of book club members who enjoy discussing a certain topic, such as literature, arts, or movies.
 **Occasional Groups**	Group members come together for a specific purpose or activity at a particular moment.	• Travelers joining a guided tour during their vacation. They share the common aim of exploring a particular attraction.
 **Random Groups**	Users share a common environment or context temporarily without explicit shared interests.	• Individuals working out at a fitness center simultaneously.

Motivation (3)

- Groups can also be categorized based on the social relationship:
  - Strong social relationship:
    - Family members planning a vacation
    - Close friends choosing a restaurant
  - Weak social relationship
    - Colleagues deciding where to go for lunch
    - Classmates selecting a project topic
  - No relationship:
    - Users collaboratively selecting songs in a shared music system

Group recommendation strategy should be adapted according to the group types or dynamics.

Motivation (4)

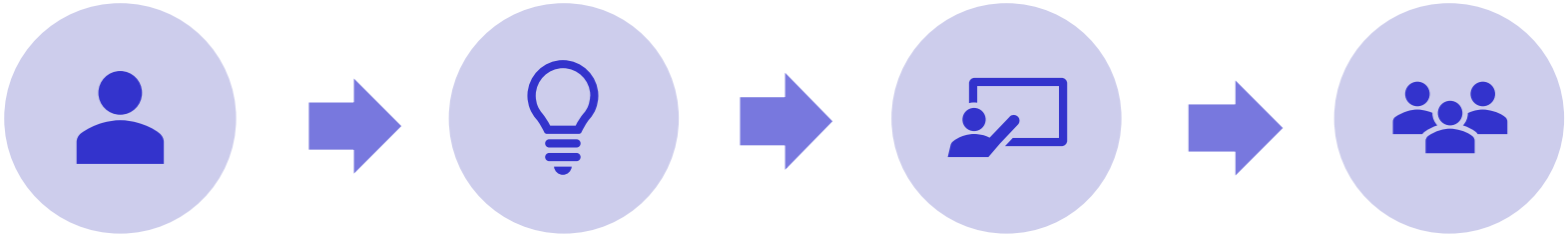
- Counteract **cognitive biases** in group decision making
  - *Cognitive biases* could lead to suboptimal group decision outcomes:
    - ***Anchoring effects*** [Stettinger2015]*:* early expressed preferences from one member influence how others evaluate their preferences
    - ***Polarization effects*** [Atas2018_Polarization]*:* group members to become more extreme in their opinions when interacting within a group
    - ***Manipulation*** [Tran2019_Manipulation]*:* some group members try to push their favourite items by strategically manipulating ratings
    - ***Emotional contagion*** [Masthoof 2011]: one member's emotions spread to others, influencing their preferences for items

Group recommender systems support group-decision making processes and improve the quality of decision outcome.

Group Recommender Systems

- **Group recommender systems**: Application of single-user recommendation approaches to groups
- Differences from recsys for individuals:
  - Different user interfaces (e.g., representing other group members)
  - Preferences of all group members influence recommendations
  - Conflicts have to be resolved
  - Consensus should be achieved
  - Fairness should be fostered
  - Cognitive biases effects impact group decision outcomes

Group Recommendation Process



1. ACQUIRING THE PREFERENCES OF INDIVIDUAL GROUP MEMBERS

2. GENERATING RECOMMENDATIONS

3. PRESENTING AND EXPLAINING RECOMMENDATIONS

4. HELPING THE GROUP TO ACHIEVE CONSENSUS AND TAKE A FINAL DECISION

Group Recommendation Process

1. Acquiring the preferences of individual group members
2. Generating recommendations
3. Presenting and explaining recommendations
4. Helping the group to achieve consensus and take a final decision

1. Acquiring Preferences (1)

- Preference acquisition in group settings does not much differ from single user scenarios

- **Implicitly** acquired preferences
  - *MusicFX* [JFM2000]: measuring time spent on listening to the songs
  - *Let's Browse* [HL1998]: analyzing the words that occur in each user's homepages

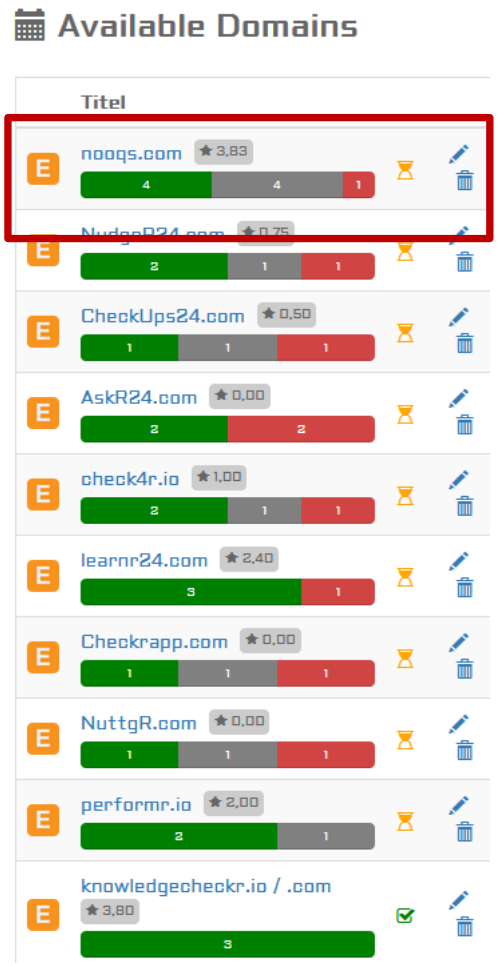
- **Explicitly** acquired preferences
  - *PocketRestaurantFinder* [JFM2002]: preferred cuisine, price, location, etc.
  - *EventHelpr* [link to the app]: pro/con arguments w.r.t. alternative items

- **Negative** preferences
  - MusicFX [JFM2000]: preferences regarding disliked genres

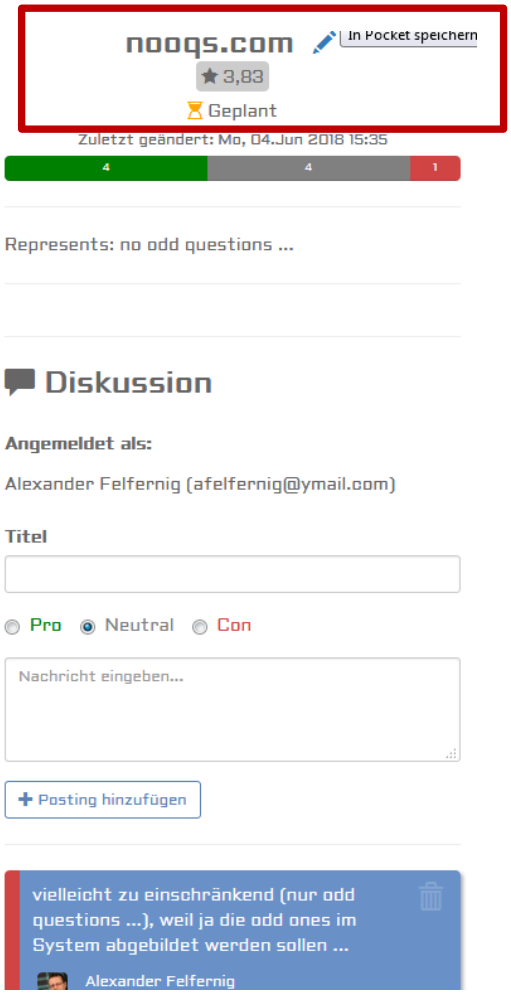
1. Acquiring Preferences (2)

- **Single-dimensional**: stars, thumbs, hearts
- **Multi-dimensional**: evaluation dimensions
  - e.g., *risk*, *profit*, *effort* in software development
- **Hard constraints vs. soft constraints**
  - **Hard constraints:** Mandatory conditions that must be satisfied (e.g., a restaurant must fit within the group's budget).
  - **Soft constraints:** Preferred conditions that improve satisfaction but can be relaxed if necessary (e.g., the group prefers outdoor seating).
- **Argumentation-based preferences**
  - pro arguments
  - con arguments

Example: EventHelpr



Task: domain name for „CheckR“



Evaluation of args: pro/con



Aggregation of args: pro/con

www.eventhelpr.com

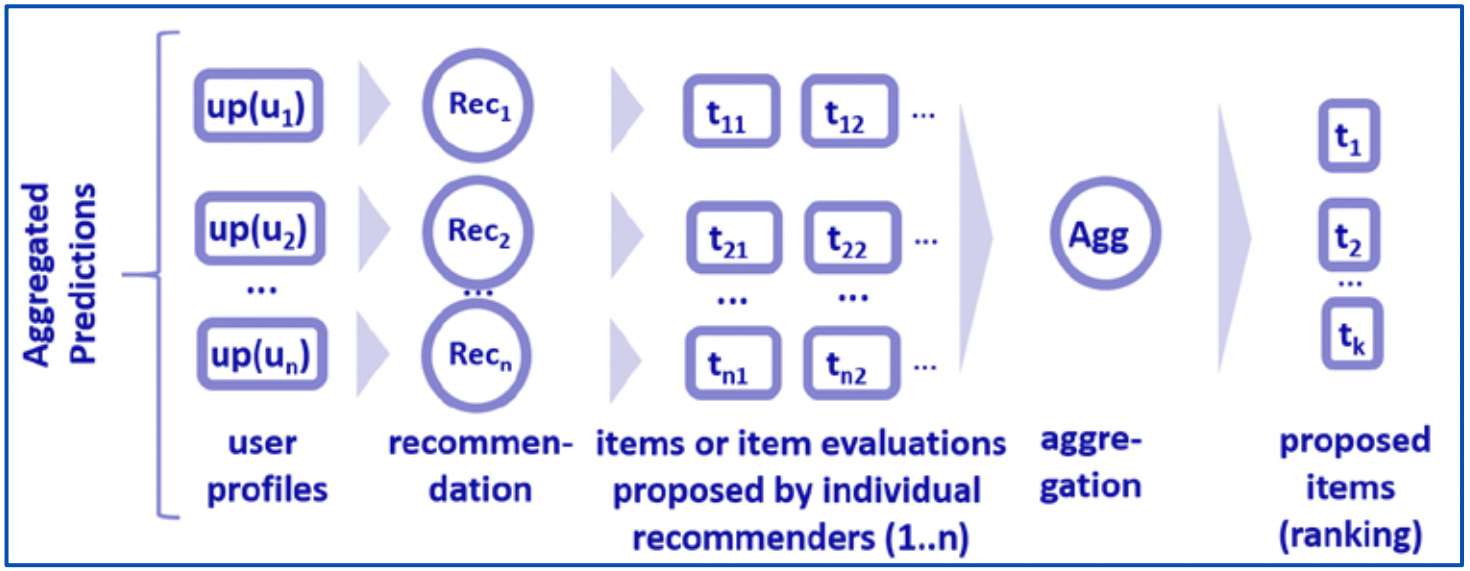
Group Recommendation Process

1. Acquiring the preferences of individual group members
2. Generating recommendations
3. Presenting and explaining recommendations
4. Helping the group to achieve consensus and take a final decision

2. Generating Recommendations

- Acquired preferences have to be aggregated
- There are two basic aggregation approaches [Felfernig2018]:
  - **Aggregated predictions:** *aggregating recommended items (or evaluations) that were generated separately for each user profile*
  - **Aggregated models:** *aggregating individual user profiles into a group profile*
- These approaches **can be applied to every basic recommendation approach**, i.e., collaborative filtering, content-based filtering, and constraint-based recommendation.

Aggregated Predictions



[Felfernig2018]

- The recommendation step is done before the aggregation step
- Items or item evaluations recommended to individual group members are aggregated into a corresponding group recommendation.

Approaches of Aggregated Predictions

Merge individual recommendation lists

- *Generate recommendations (items) separately for each group member*
- *Merge all recommended items into a set of candidate items*
- *Items disliked by one member should not be filtered out → to preserve diverse preferences before group discussion*
  - *Example: Restaurant A liked by u_1 should not be excluded although u_2 does not like it.*
- *No ranking is provided → the group has to make the final decision together*

Aggregating Individual Predictions

- Predict each member's preference for candidate items
- Aggregate these predictions into group-level scores
- Produce a ranked list of candidate items
- The system determines the final group recommendation

[Felfernig2018]

Merge individual recommendation lists

item	name	rating predictions			
		u_1	u_2	u_3	u_4
t_1	Vienna	5.0 (5)	3.5 (1)	1.0 (0)	4.5 (5)
t_2	Paris	2.5 (0)	4.0 (3)	3.0 (1)	2.0 (0),
t_3	New York	4.9 (4)	3.8 (2)	4.0 (3)	3.3 (3)
t_4	Barcelona	3.1 (1)	5.0 (5)	4.2 (4)	2.4 (1)
t_5	London	4.0 (2)	4.3 (4)	3.3 (2)	4.1 (4)
t_6	Tokyo	4.5 (3)	4.1 (3)	5.0 (5)	3.2 (2)

candidate items for group
Vienna
Barcelona
Tokyo

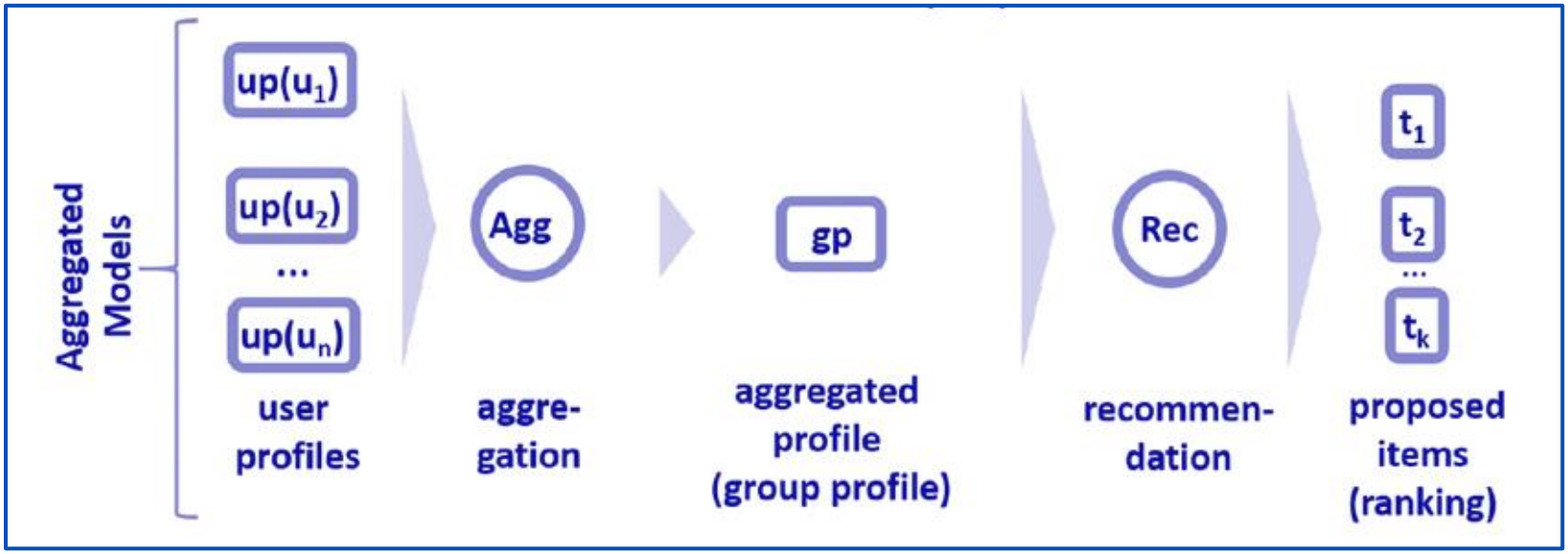
the group jointly makes the final decision

Aggregating Individual Predictions

item	name	rating predictions				Agg. Func.
		u_1	u_2	u_3	u_4	AVERAGE (AVV)
t_1	Vienna	5.0 (5)	3.5 (1)	1.0 (0)	4.5 (5)	3.5 (1)
t_2	Paris	2.5 (0)	4.0 (3)	3.0 (1)	2.0 (0)	2.9 (0)
t_3	New York	4.9 (4)	3.8 (2)	4.0 (3)	3.3 (3)	4.0 (4)
t_4	Barcelona	3.1 (1)	5.0 (5)	4.2 (4)	2.4 (1)	3.7 (2)
t_5	London	4.0 (2)	4.3 (4)	3.3 (2)	4.1 (4)	3.9 (3)
t_6	Tokyo	4.5 (3)	4.1 (3)	5.0 (5)	3.2 (2)	4.2 (5) ✅

Top-3
Tokyo
New York
London

Aggregated Models (Group Profiles)



[Felfernig2018]

- This approach first aggregates multiple user preferences into a single group profile.
- The system then generates recommendations based on the group profile.

Social Choice-based Preference Aggregation Functions

- no optimal way to aggregate recommendation lists
- corresponding approximations have to be used:
  - **social choice-based preference aggregation functions**
  - generate a recommendation considering **as far as possible** the individual preferences of group members

Average Vote (AVV)

$$AVV(t) = \frac{\sum_{u \in U} eval(u, t)}{|U|}$$

Item (t)	Martin (u_1)	Alex (u_2)	Chris (u_3)	Joe (u_4)
Katz's Delicatessen	5	3	5	4
Joe's Pizza	3	3	5	3
Shake Shack	5	3	3	3
Le Bernardin	4	3	4	4

AVV (t)
4.25
3.5
3.5
2.75



Majority Vote (MAJV)

$$MAJV(t) = argmax_{d \in D} |\{u \in U | eval(u,t) = d\}|$$

D is the set of possible ratings/evaluations

Item (t)	Martin (u_1)	Alex (u_2)	Chris (u_3)	Joe (u_4)	MAJV (t)
Katz's Delicatessen	5	3	5	4	**5**
Joe's Pizza	3	3	5	3	3
Shake Shack	5	3	3	3	3
Le Bernardin	4	3	4	4	4





Least Misery (LEMI)

$$LEMI(t) = min_{u \in U}\,(\mathrm{eval}(\mathrm{u}, \mathrm{t}))$$

Item (t)	Martin (u_1)	Alex (u_2)	Chris (u_3)	Joe (u_4)
Katz's Delicatessen	5	3	5	4
Joe's Pizza	3	3	5	3
Shake Shack	5	3	3	3
Le Bernardin	4	3	4	4



LEMI (t)
3
3
3
3

Most Pleasure (MOPL)

$$MOPL(t) = max_{u \in U}(eval(u, t))$$

Item (t)	Martin (u_1)	Alex (u_2)	Chris (u_3)	Joe (u_4)
Katz's Delicatessen	5	3	5	4
Joe's Pizza	3	3	5	3
Shake Shack	5	3	3	3
Le Bernardin	4	3	4	4



MOPL (t)
5
5
5
4

Group Distance (GRDI)

$$GRDI(t) = argmin_{d \in D} \sum_{u \in U} |eval(u,t) - d|$$

distance(5) = 2 + 0 + 1 = 3
distance(3) = 2 + 2 + 1 = 5
distance(4) = 1 + 1 + 1 = 3

Item (t)	Martin (u_1)	Alex (u_2)	Chris (u_3)	Joe (u_4)
Katz's Delicatessen	5	3	5	4
Joe's Pizza	3	3	5	3
Shake Shack	5	3	3	3
Le Bernardin	4	3	4	4

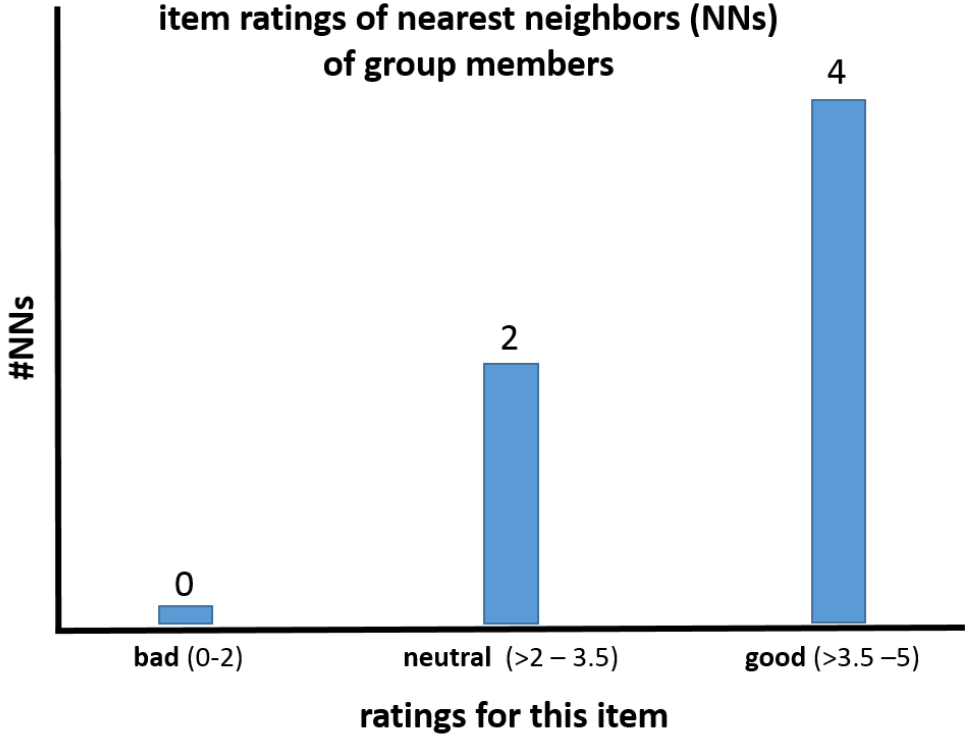
2 1 0

GRDI (t)
5
3
3
4

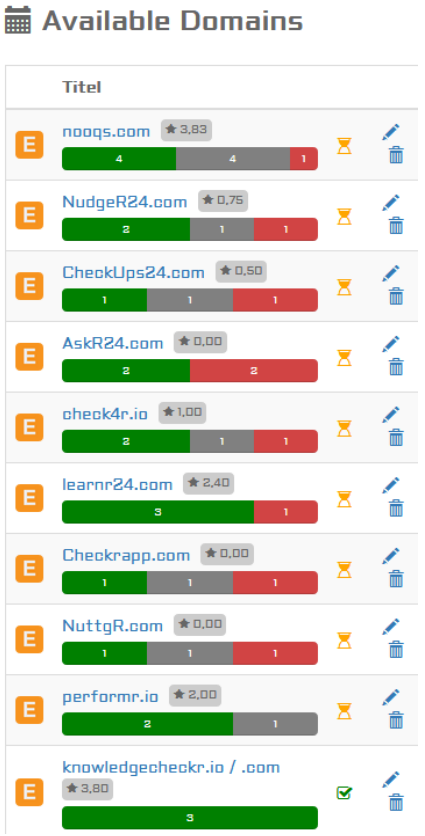
3. Presenting & Explaining Recommendations

- Explanations can have **different goals**, for example, transparency, trust, domain understanding, satisfaction, and persuasion.
- **Different approaches**:
  - textual explanations (e.g., “similar group members also purchased X”)
  - graphical explanations (e.g., via bar charts)
- **Explanations differ** depending on:
  - the used basic recommendation approach
  - the used aggregation strategy
  - social aspects (e.g., fairness)

Further Example Explanations



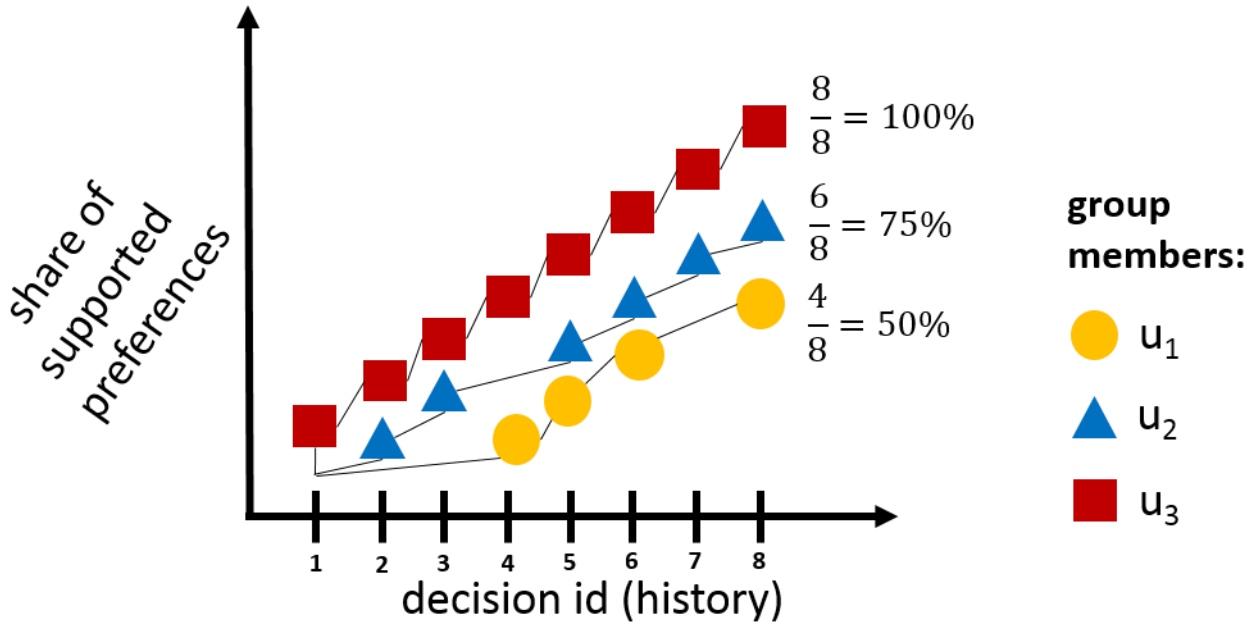
How nearest neighbors rated a recommended item.



How group members evaluated an item on the basis of arguments.

Example: Explaining Fairness Aspects (1)

- Preference support in the past should be taken into account in future recommendations …



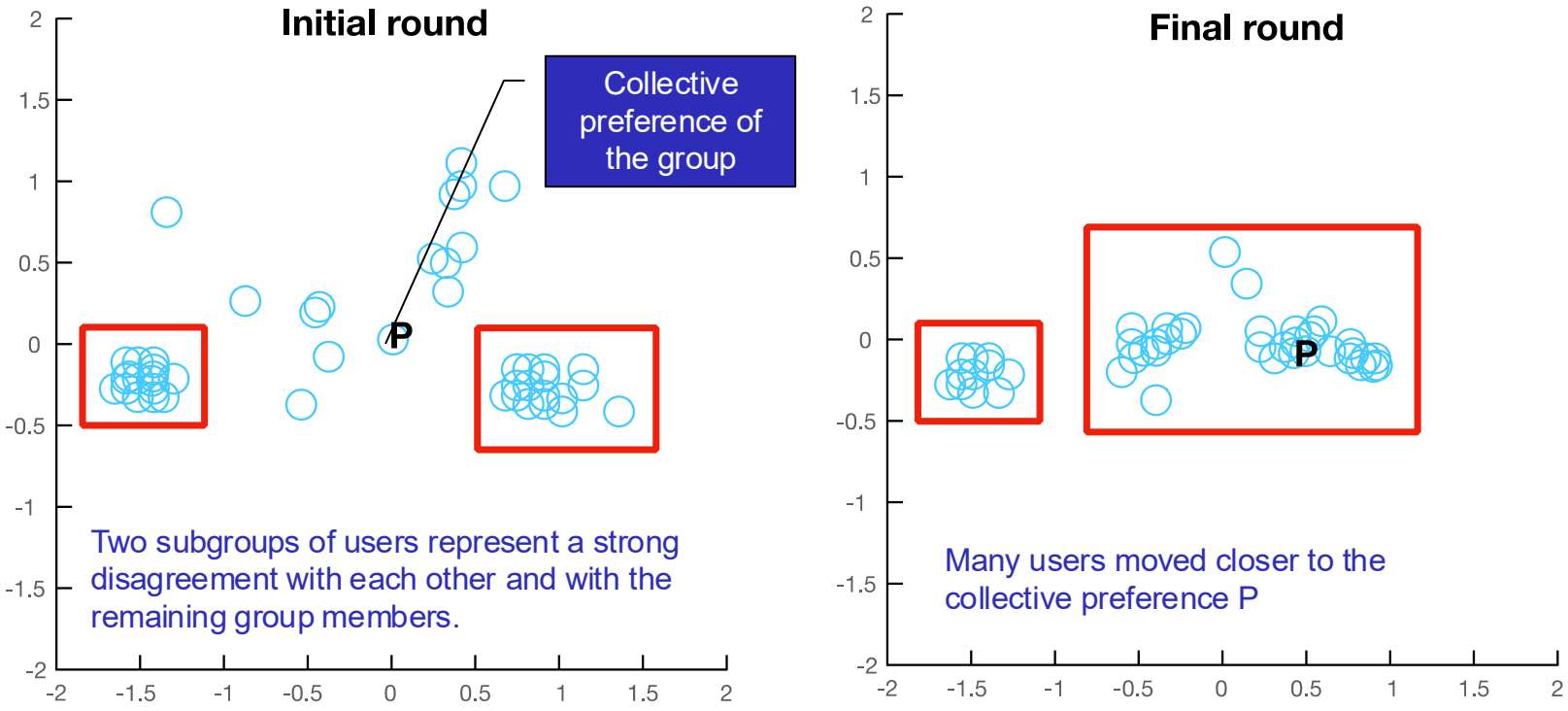
[Felfernig et al., 2018]

4. Consensus “Scenarios”

- From **recommendation into action**
  - recommendation = decision
  - e.g., restaurant with the highest avg. voting (agreement needed beforehand)
  - e.g., recommended music is automatically played
- From **recommendation into discussion**
  - recommendation = input to discussion phase
  - E.g., Internet domain name with the highest support in terms of pro-arguments and an “acceptable” amount of con-arguments
- Final decision by **responsible persons**
  - recommendation = input to discussion, however, person A decides …
  - e.g., A: the new company car of type X will be purchased

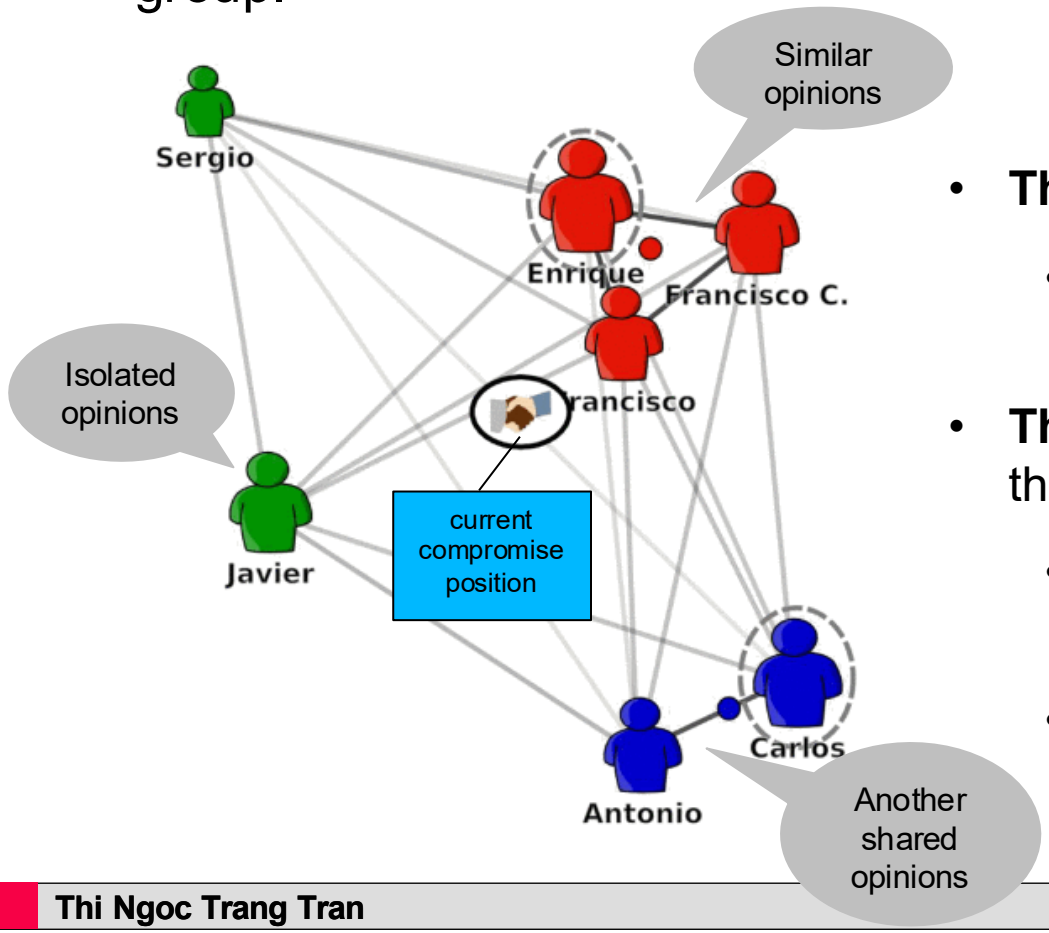
Visualization Methods Supporting the Consensus-Making Process (1)

- Set-based graphs **[Palomares2014]** represent the consensus state of a group in the initial round and in the final round of the consensus-achieving process



Visualization Methods Supporting the Consensus-Making Process (2)

- Node-link graphs **[Alonso2007]** visualize the preference conflicts in the group.



- **The nodes** represent group members
  - Smaller node: more contradictory preferences
- **The links** show conflicts between them.
  - Longer link: higher conflict between group members
  - The link between Sergio and Antonio indicates a high conflict between these two users

Further Aspects in Group Recommendation

- Achieving fairness (e.g., in repeated decisions) **[Felfernig2018, Tran2019_Explanations]**
- Accelerating consensus making **[Tran2019_Explanations, Tran2023]**
- Treating group members differently where appropriate (approach: “liquid democracy”) **[Atas2018]**
- Explanations **[Felfernig2018, Tran2019_Explanations]**
- Biases:
  - Avoiding biases (e.g., anchoring) **[Felfernig2018, Stettinger2015]**
  - Discouraging manipulation of the recommendation mechanism (approach: transparent UIs) **[Tran2019_Manipulation]**

Increasing Fairness and Accelerating Consensus Making

Explaining aggregation function

Item X has been recommended to the group since most group members like it.

Explaining aggregation function + decision history

Item X has been recommended to the group since most group members like it. **This decision supports the preferences of users u_a, u_b, and u_c who were treated less favorably in the last n decisions.**

Explaining aggregation function + future decision plan

Item X has been recommended to the group since most group members like it. **The preference of user u_a seems not to be considered in this decision. Therefore, all group members agreed that he/she will have a higher priority in the next decision.**

increase fairness perception and accelerate consensus-making process

[Tran2019_Explanations]

Liquid Democracy in Group Decision Making

Traditional group evaluation

Users → Evaluate requirements → Recommendation

Issues:

- some group members do not know enough about certain requirements
- treat both group members and expert, then same
- provide incomplete or uncertain evaluations

Liquid Democracy Approach

Users
- evaluate directly
- delegate to experts

↓

Expert-supported evaluation

↓

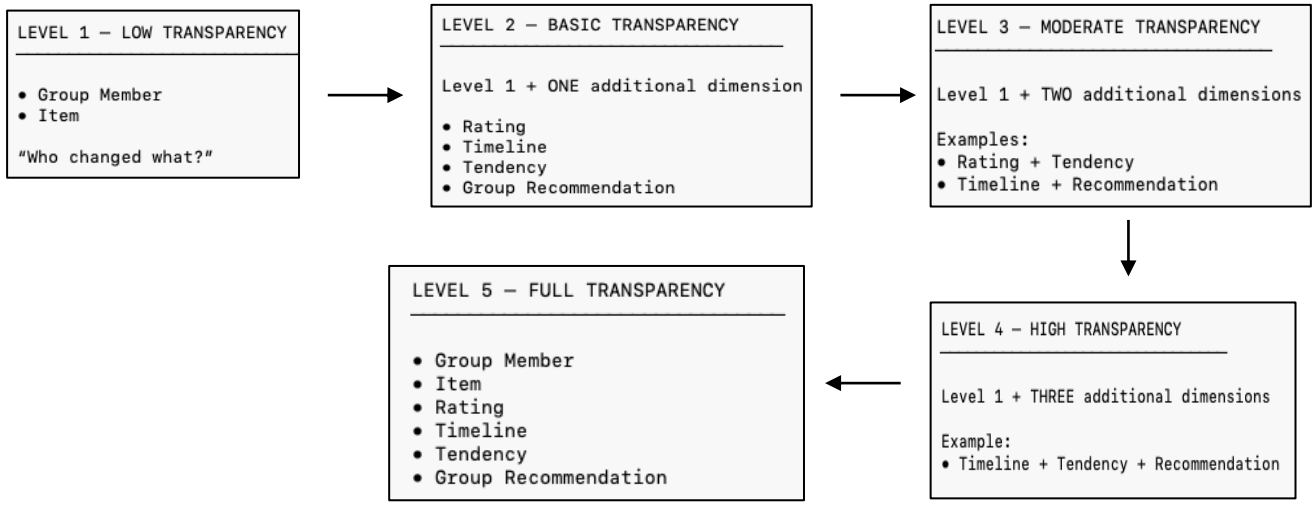
Better recommendations

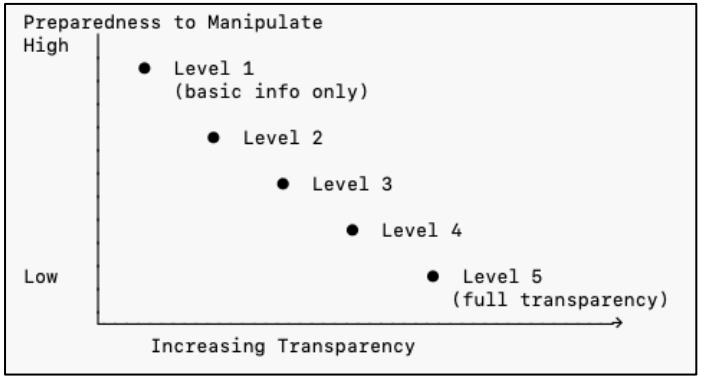
Expert-aware Utility Calculation

- **Utility factors**: risk, effort, product, and expert importance
- **MAUT**: (Stakeholders evaluations + Expert evaluations + Expertise level) → Utility calculation → Better release planning
  - **Experts evaluations receive higher influence/weight**
- **Outcome**: Higher utility → Higher priority

[Atas2018]

Manipulation Counteraction





[Tran2019_Manipulation]

Higher transparency → lower manipulation tendency

- Full transparency produced the lowest willingness to manipulate
- 66.67% of participants avoided manipulation under full transparency
- Users became more cautious when rating adaptations were fully visible

[Tran2019_Manipulation]

Ongoing and Future Research

- **Beyond Item Recommendation**: usage of recommenders to stimulate specific decision behavior
- Development of mechanisms to **counteract decision biases/effects** (e.g., "*anchoring*", "*emotional contagion*", "*conformity*" **[Masthoff2011]**)
  - *Emotional contagion*: depends on your personality and relationship with others. *You are more likely to be contaged by somebody you love.*
  - *Conformity*: *You want to be part of the group and express an opinion like the rest of the group even though inside you still belief differently.*
- **Psychology-informed** recommender systems
- **Meta-level decision strategies**
- Group **recommender systems in SE** and for **complex items**

Exercise

- Choose a domain and define your own group recommendation task
- Define a group of five persons and assume their rating on at least 4 alternative items
- Choose the from your point of view most relevant aggregation strategy
- Apply the chosen strategy to your setting and identify the most appropriate item for the group
- Note: no submission („Abgabe“) needed!

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Thank You!